Retention analysis

Brazilian E-Commerce
Public Dataset by Olist

Justification of Retention Analysis

Dashboard showed that new customers share is about 98% even after two years after launch.

This brought me to the idea to analyze, how exactly retention looks like.

Executive summary

- 97% customers make only one order
- Repeat AOV is 8.4% lower than first orders
- Low retention is driven by product mix (durable goods)
- Retention focus should be selective, not global

Main results from data analysis

Distribution of orders count per customer

During the observed time period:

- 97% of all customers made one order
- 2.8% made exactly two orders
- only 0.2% made more than two orders

Time to the repeat order

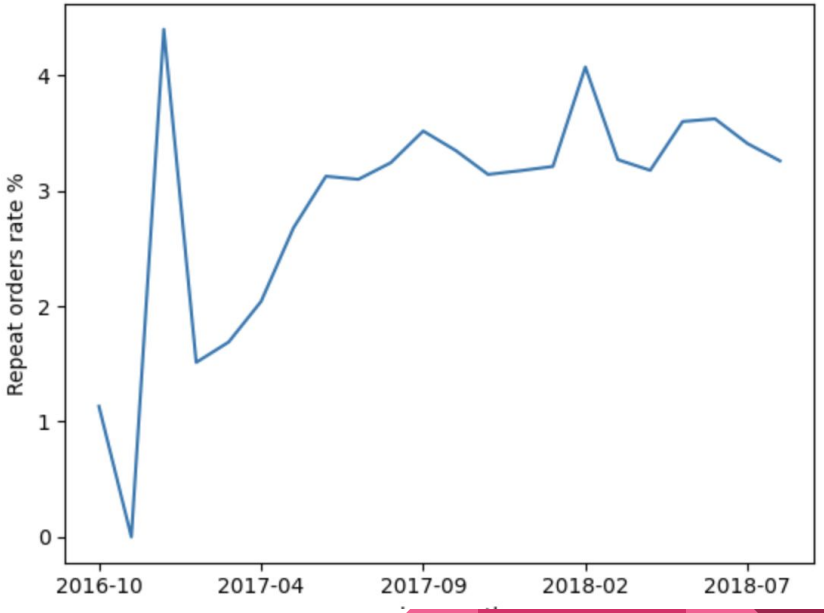
Since the data shows only a relatively short period of time, it makes sense to check, what time does it take for customers to make a repeat order.

- average time is 79 days
- but median is 29 days
- with time this number only grows, which is logical, considering we have data starting from the date of launch

So the time span is enough to get repeat orders.

Repeat orders share from all orders

Repeat share stabilizes around 3% with time, suggesting that most customers interact with the marketplace as one-time buyers.



Conclusion

Business growth is primarily driven by new customer acquisition rather than retention.

Comparing new vs repeat orders

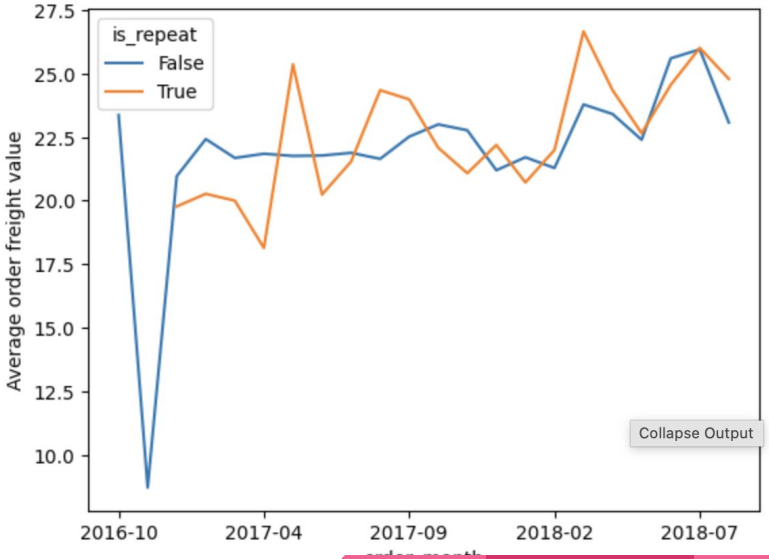
Overall Repeat vs First order

Data shows that AOV of repeat orders is 8.4% smaller than AOV of first orders, while freight value is 1.6% larger.

Order type	AOV	Freight value	Orders Count
First	160.29	22.77	93 357
Repeat	146.92	23.14	3 120

Repeat vs First order monthly comparison

Those values do not change much with time.



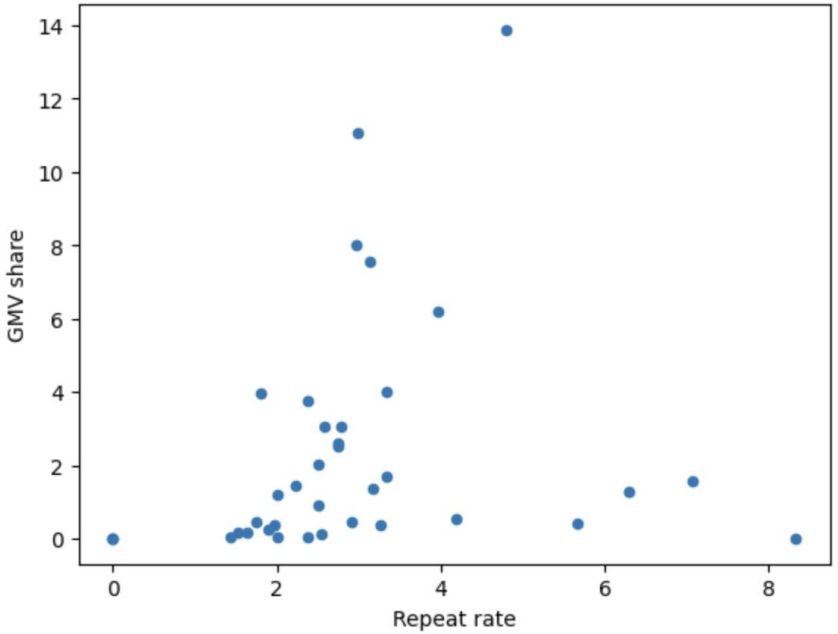
Conclusion

Repeat orders contribute only ~3% of monthly orders and have lower AOV, making their current impact on GMV marginal.

Item category analysis

Categories repeat rate and GMV share

Distribution of categories shows that the most profitable categories have low repeat rate, and the ones with high repeat rate have little impact on GMV.



High GMV categories

Top categories by GMV are low-frequency by nature (Furniture, Electronics, ...)

Product category	GMV share (%)	Repeat rate (%)
Furniture	13.9	4.8
Electronics	11.1	3.0
Health & Beauty	8.0	3.0

Top repeat categories

High-repeat categories contribute negligible GMV

Product category	GMV share (%)	Repeat rate (%)
Arts and Craftsmanship	0.0	8.3
Home Appliances	1.6	7.1
Fashion	1.3	6.3

Potential high-repeat categories

The following consumable categories have a potential to contribute more into GMV by increasing their retention rate - GMV share isn't too small while repeat share could be larger

Product category	GMV share (%)	Repeat rate (%)
Baby	2.6	2.7
Pet shop	1.4	3.2
Toys	3.1	2.8

Conclusion

Low retention appears structural rather than operational

Analysis results

Given current data, acquisition appears to be the primary GMV driver. Improving retention may have limited impact on GMV. However, evaluating CAC and LTV is required before strategic prioritization. Potentially testing targeted retention initiatives in consumable categories (e.g., Baby, Pet, Toys).

Limitations & Next Steps

The analysis does not include customer acquisition cost (CAC).

Repeat customers may be more profitable despite slightly lower AOV if acquisition costs are significant.

Further analysis should incorporate marketing spend data to evaluate true customer lifetime value.